

7. Diffusions: Basic Properties

7.1 The Markov Property

Suppose we want to describe the motion of a small particle suspended in a moving liquid, subject to random molecular bombardments. If $b(t, x) \in \mathbf{R}^3$ is the velocity of the fluid at the point x at time t , then a reasonable mathematical model for the position X_t of the particle at time t would be a stochastic differential equation of the form

$$\frac{dX_t}{dt} = b(t, X_t) + \sigma(t, X_t)W_t, \quad (7.1.1)$$

where $W_t \in \mathbf{R}^3$ denotes “white noise” and $\sigma(t, x) \in \mathbf{R}^{3 \times 3}$. The Itô interpretation of this equation is

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t, \quad (7.1.2)$$

where B_t is 3-dimensional Brownian motion, and similarly (with a correction term added to b) for the Stratonovich interpretation (see (6.1.3)).

In a stochastic differential equation of the form

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t, \quad (7.1.3)$$

where $X_t \in \mathbf{R}^n$, $b(t, x) \in \mathbf{R}^n$, $\sigma(t, x) \in \mathbf{R}^{n \times m}$ and B_t is m -dimensional Brownian motion, we will call b the *drift coefficient* and σ – or sometimes $\frac{1}{2}\sigma\sigma^T$ – the *diffusion coefficient* (see Theorem 7.3.3).

Thus the solution of a stochastic differential equation may be thought of as the mathematical description of the motion of a small particle in a moving fluid: Therefore such stochastic processes are called *(Itô) diffusions*.

In this chapter we establish some of the most basic properties and results about Itô diffusions:

- 7.1 The Markov property.
- 7.2 The strong Markov property.
- 7.3 The generator A of X_t expressed in terms of b and σ .
- 7.4 The Dynkin formula.
- 7.5 The characteristic operator.

This will give us the necessary background for the applications in the remaining chapters.

Definition 7.1.1. A (time-homogeneous) Itô diffusion is a stochastic process $X_t(\omega) = X(t, \omega): [0, \infty) \times \Omega \rightarrow \mathbf{R}^n$ satisfying a stochastic differential equation of the form

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t, \quad t \geq s; \quad X_s = x \quad (7.1.4)$$

where B_t is m -dimensional Brownian motion and $b: \mathbf{R}^n \rightarrow \mathbf{R}^n$, $\sigma: \mathbf{R}^n \rightarrow \mathbf{R}^{n \times m}$ satisfy the conditions in Theorem 5.2.1, which in this case simplify to:

$$|b(x) - b(y)| + |\sigma(x) - \sigma(y)| \leq D|x - y|; \quad x, y \in \mathbf{R}^n, \quad (7.1.5)$$

where $|\sigma|^2 = \sum |\sigma_{ij}|^2$.

We will denote the (unique) solution of (7.1.4) by $X_t = X_t^{s,x}$; $t \geq s$. If $s = 0$ we write X_t^x for $X_t^{0,x}$. Note that we have assumed in (7.1.4) that b and σ do not depend on t but on x only. We shall see later (Chapters 10, 11) that the general case can be reduced to this situation. The resulting process $X_t(\omega)$ will have the property of being *time-homogeneous*, in the following sense:

Note that

$$\begin{aligned} X_{s+h}^{s,x} &= x + \int_s^{s+h} b(X_u^{s,x})du + \int_s^{s+h} \sigma(X_u^{s,x})dB_u \\ &= x + \int_0^h b(X_{s+v}^{s,x})dv + \int_0^h \sigma(X_{s+v}^{s,x})d\tilde{B}_v, \quad (u = s + v) \end{aligned} \quad (7.1.6)$$

where $\tilde{B}_v = B_{s+v} - B_s$; $v \geq 0$. (See Exercise 2.12). On the other hand of course

$$X_h^{0,x} = x + \int_0^h b(X_v^{0,x})dv + \int_0^h \sigma(X_v^{0,x})dB_v.$$

Since $\{\tilde{B}_v\}_{v \geq 0}$ and $\{B_v\}_{v \geq 0}$ have the same P^0 -distributions, it follows by weak uniqueness (Lemma 5.3.1) of the solution of the stochastic differential equation

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t; \quad X_0 = x$$

that

$$\{X_{s+h}^{s,x}\}_{h \geq 0} \quad \text{and} \quad \{X_h^{0,x}\}_{h \geq 0}$$

have the same P^0 -distributions, i.e. $\{X_t\}_{t \geq 0}$ is *time-homogeneous*.

We now introduce the probability laws Q^x of $\{X_t\}_{t \geq 0}$, for $x \in \mathbf{R}^n$. Intuitively, Q^x gives the distribution of $\{X_t\}_{t \geq 0}$ assuming that $X_0 = x$. To express

this mathematically, we let $\mathcal{M}_\infty$ be the σ -algebra (of subsets of Ω) generated by the random variables $\omega \rightarrow X_t(\omega) = X_t^y(\omega)$, where $t \geq 0$, $y \in \mathbf{R}^n$.

Define Q^x on the members of $\mathcal{M}$ by

$$Q^x[X_{t_1} \in E_1, \dots, X_{t_k} \in E_k] = P^0[X_{t_1}^x \in E_1, \dots, X_{t_k}^x \in E_k] \quad (7.1.7)$$

where $E_i \subset \mathbf{R}^n$ are Borel sets; $1 \leq i \leq k$.

As before we let $\mathcal{F}_t^{(m)}$ be the σ -algebra generated by $\{B_r; r \leq t\}$. Similarly we let $\mathcal{M}_t$ be the σ -algebra generated by $\{X_r; r \leq t\}$. We have established earlier (see Theorem 5.2.1) that X_t is measurable with respect to $\mathcal{F}_t^{(m)}$, so $\mathcal{M}_t \subseteq \mathcal{F}_t^{(m)}$.

We now prove that X_t satisfies the important *Markov property*: The future behaviour of the process given what has happened up to time t is the same as the behaviour obtained when starting the process at X_t . The precise mathematical formulation of this is the following:

Theorem 7.1.2 (The Markov property for Itô diffusions).

Let f be a bounded Borel function from $\mathbf{R}^n$ to $\mathbf{R}$. Then, for $t, h \geq 0$

$$E^x[f(X_{t+h})|\mathcal{F}_t^{(m)}]_{(\omega)} = E^{X_t(\omega)}[f(X_h)] . \quad (7.1.8)$$

(See Appendix B for definition and basic properties of conditional expectation). Here and in the following E^x denotes the expectation w.r.t. the probability measure Q^x . Thus $E^y[f(X_h)]$ means $E[f(X_h^y)]$, where E denotes the expectation w.r.t. the measure P^0 . The right hand side means the function $E^y[f(X_h)]$ evaluated at $y = X_t(\omega)$.

Proof. Since, for $r \geq t$,

$$X_r(\omega) = X_t(\omega) + \int_t^r b(X_u)du + \int_t^r \sigma(X_u)dB_u ,$$

we have by uniqueness

$$X_r(\omega) = X_r^{t, X_t}(\omega) .$$

In other words, if we define

$$F(x, t, r, \omega) = X_r^{t, x}(\omega) \quad \text{for } r \geq t ,$$

we have

$$X_r(\omega) = F(X_t, t, r, \omega); \quad r \geq t . \quad (7.1.9)$$

Note that $\omega \rightarrow F(x, t, r, \omega)$ is independent of $\mathcal{F}_t^{(m)}$. Using (7.1.9) we may rewrite (7.1.8) as

$$E[f(F(X_t, t, t+h, \omega))|\mathcal{F}_t^{(m)}] = E[f(F(x, 0, h, \omega))]_{x=X_t} . \quad (7.1.10)$$

Put $g(x, \omega) = f \circ F(x, t, t + h, \omega)$. Then $(x, \omega) \rightarrow g(x, \omega)$ is measurable. (See Exercise 7.6). Hence we can approximate g pointwise boundedly by functions on the form

$$\sum_{k=1}^m \phi_k(x) \psi_k(\omega) .$$

Using the properties of conditional expectation (see Appendix B) we get

$$\begin{aligned} E[g(X_t, \omega) | \mathcal{F}_t^{(m)}] &= E \left[\lim \sum \phi_k(X_t) \psi_k(\omega) | \mathcal{F}_t^{(m)} \right] \\ &= \lim \sum \phi_k(X_t) \cdot E[\psi_k(\omega) | \mathcal{F}_t^{(m)}] \\ &= \lim \sum E[\phi_k(y) \psi_k(\omega) | \mathcal{F}_t^{(m)}]_{y=X_t} \\ &= E[g(y, \omega) | \mathcal{F}_t^{(m)}]_{y=X_t} = E[g(y, \omega)]_{y=X_t} . \end{aligned}$$

Therefore, since $\{X_t\}$ is time-homogeneous,

$$\begin{aligned} E[f(F(X_t, t, t + h, \omega)) | \mathcal{F}_t^{(m)}] &= E[f(F(y, t, t + h, \omega))]_{y=X_t} \\ &= E[f(F(y, 0, h, \omega))]_{y=X_t} \end{aligned}$$

which is (7.1.10). $\square$

Remark. Theorem 7.1.2 states that X_t is a Markov process w.r.t. the family of σ -algebras $\{\mathcal{F}_t^{(m)}\}_{t \geq 0}$. Note that since $\mathcal{M}_t \subseteq \mathcal{F}_t^{(m)}$ this implies that X_t is also a Markov process w.r.t. the σ -algebras $\{\mathcal{M}_t\}_{t \geq 0}$. This follows from Theorem B.3 and Theorem B.2 c) (Appendix B):

$$\begin{aligned} E^x[f(X_{t+h}) | \mathcal{M}_t] &= E^x[E^x[f(X_{t+h}) | \mathcal{F}_t^{(m)}] | \mathcal{M}_t] \\ &= E^x[E^{X_t}[f(X_h)] | \mathcal{M}_t] = E^{X_t}[f(X_h)] \end{aligned}$$

since $E^{X_t}[f(X_h)]$ is $\mathcal{M}_t$ -measurable.

7.2 The Strong Markov Property

Roughly, the strong Markov property states that a relation of the form (7.1.8) continues to hold if the time t is replaced by a random time $\tau(\omega)$ of a more general type called *stopping time* (or *Markov time*):

Definition 7.2.1. Let $\{\mathcal{N}_t\}$ be an increasing family of σ -algebras (of subsets of Ω). A function $\tau: \Omega \rightarrow [0, \infty]$ is called a (strict) stopping time w.r.t. $\{\mathcal{N}_t\}$ if

$$\{\omega; \tau(\omega) \leq t\} \in \mathcal{N}_t, \quad \text{for all } t \geq 0 .$$

In other words, it should be possible to decide whether or not $\tau \leq t$ has occurred on the basis of the knowledge of $\mathcal{N}_t$.

Example 7.2.2. Let $U \subset \mathbf{R}^n$ be open. Then the *first exit time*

$$\tau_U := \inf\{t > 0; X_t \notin U\}$$

is a stopping time w.r.t. $\{\mathcal{M}_t\}$, since

$$\{\omega; \tau_U \leq t\} = \bigcap_m \bigcup_{\substack{r \in \mathbf{Q} \\ r < t}} \{\omega; X_r \notin K_m\} \in \mathcal{M}_t$$

where $\{K_m\}$ is an increasing sequence of closed sets such that $U = \bigcup_m K_m$.

More generally, if $H \subset \mathbf{R}^n$ is any set we define the *first exit time from H* , τ_H , as follows

$$\tau_H = \inf\{t > 0; X_t \notin H\}.$$

If we include the sets of measure 0 in $\mathcal{M}_t$ (which we do) then the family $\{\mathcal{M}_t\}$ is right-continuous i.e. $\mathcal{M}_t = \mathcal{M}_{t+}$, where $\mathcal{M}_{t+} = \bigcap_{s>t} \mathcal{M}_s$ (see Chung (1982, Theorem 2.3.4., p. 61)) and therefore τ_H is a stopping time for any Borel set H (see Dynkin (1965 II, 4.5.C.e.), p. 111)).

Definition 7.2.3. Let τ be a stopping time w.r.t. $\{\mathcal{N}_t\}$ and let $\mathcal{N}_\infty$ be the smallest σ -algebra containing $\mathcal{N}_t$ for all $t \geq 0$. Then the σ -algebra $\mathcal{N}_\tau$ consists of all sets $N \in \mathcal{N}_\infty$ such that

$$N \cap \{\tau \leq t\} \in \mathcal{N}_t \quad \text{for all } t \geq 0.$$

In the case when $\mathcal{N}_t = \mathcal{M}_t$, an alternative and more intuitive description is:

$$\mathcal{M}_\tau = \text{the } \sigma\text{-algebra generated by } \{X_{\min(s, \tau)}; s \geq 0\}. \quad (7.2.1)$$

(See Rao (1977, p. 2.15) or Stroock and Varadhan (1979, Lemma 1.3.3, p. 33).) Similarly, if $\mathcal{N}_t = \mathcal{F}_t^{(m)}$, we get

$$\mathcal{F}_\tau^{(m)} = \text{the } \sigma\text{-algebra generated by } \{B_{s \wedge \tau}; s \geq 0\}.$$

Theorem 7.2.4 (The strong Markov property for Itô diffusions).

Let f be a bounded Borel function on $\mathbf{R}^n$, τ a stopping time w.r.t. $\mathcal{F}_t^{(m)}$, $\tau < \infty$ a.s. Then

$$E^x[f(X_{\tau+h}) | \mathcal{F}_\tau^{(m)}] = E^{X_\tau}[f(X_h)] \quad \text{for all } h \geq 0. \quad (7.2.2)$$

Proof. We try to imitate the proof of the Markov property (Theorem 7.1.2). For a.a. ω we have that $X_\tau^{\tau, x}(\omega)$ satisfies

$$X_{\tau+h}^{\tau, x} = x + \int_\tau^{\tau+h} b(X_u^{\tau, x}) du + \int_\tau^{\tau+h} \sigma(X_u^{\tau, x}) dB_u.$$

By the strong Markov property for Brownian motion (Gihman and Skorohod (1974a, p. 30)) the process

$$\tilde{B}_v = B_{\tau+v} - B_\tau ; \quad v \geq 0$$

is again a Brownian motion and independent of $\mathcal{F}_\tau^{(m)}$. Therefore

$$X_{\tau+h}^{\tau,x} = x + \int_0^h b(X_{\tau+v}^{\tau,x}) dv + \int_0^h \sigma(X_{\tau+v}^{\tau,x}) d\tilde{B}_v .$$

Hence $\{X_{\tau+h}^{\tau,x}\}_{h \geq 0}$ must coincide a.e. with the strongly unique (see (5.2.8)) solution Y_h of the equation

$$Y_h = x + \int_0^h b(Y_v) dv + \int_0^h \sigma(Y_v) d\tilde{B}_v .$$

Since $\{Y_h\}_{h \geq 0}$ is independent of $\mathcal{F}_\tau^{(m)}$, $\{X_{\tau+h}^{\tau,x}\}$ must be independent also. Moreover, by weak uniqueness (Lemma 5.3.1) we conclude that

$$\{Y_h\}_{h \geq 0} , \text{ and hence } \{X_{\tau+h}^{\tau,x}\}_{h \geq 0}, \text{ has the same law as } \{X_h^{0,x}\}_{h \geq 0} . \quad (7.2.3)$$

Put

$$F(x, t, r, \omega) = X_r^{t,x}(\omega) \quad \text{for } r \geq t .$$

Then (7.2.2) can be written

$$E[f(F(x, 0, \tau + h, \omega)) | \mathcal{F}_\tau^{(m)}] = E[f(F(x, 0, h, \omega))]_{x=X_\tau^{0,x}} .$$

Now, with $X_t = X_t^{0,x}$,

$$\begin{aligned} F(x, 0, \tau + h, \omega) &= X_{\tau+h}(\omega) = x + \int_0^{\tau+h} b(X_s) ds + \int_0^{\tau+h} \sigma(X_s) dB_s \\ &= x + \int_0^\tau b(X_s) ds + \int_0^\tau \sigma(X_s) dB_s + \int_\tau^{\tau+h} b(X_s) ds + \int_\tau^{\tau+h} \sigma(X_s) dB_s \\ &= X_\tau + \int_\tau^{\tau+h} b(X_s) ds + \int_\tau^{\tau+h} \sigma(X_s) dB_s \\ &= F(X_\tau, \tau, \tau + h, \omega) . \end{aligned}$$

Hence (7.2.2) gets the form

$$E[f(F(X_\tau, \tau, \tau + h, \omega)) | \mathcal{F}_\tau^{(m)}] = E[f(F(x, 0, h, \omega))]_{x=X_\tau} .$$

Put $g(x, t, r, \omega) = f(F(x, t, r, \omega))$. As in the proof of Theorem 7.1.2 we may assume that g has the form

$$g(x, t, r, \omega) = \sum_k \phi_k(x) \psi_k(t, r, \omega) .$$

Then, since $X_{\tau+h}^{\tau, x}$ is independent of $\mathcal{F}_\tau^{(m)}$ we get, using (7.2.3)

$$\begin{aligned} E[g(X_\tau, \tau, \tau+h, \omega) | \mathcal{F}_\tau^{(m)}] &= \sum_k E[\phi_k(X_\tau) \psi_k(\tau, \tau+h, \omega) | \mathcal{F}_\tau^{(m)}] \\ &= \sum_k \phi_k(X_\tau) E[\psi_k(\tau, \tau+h, \omega) | \mathcal{F}_\tau^{(m)}] = \sum_k E[\phi_k(x) \psi_k(\tau, \tau+h, \omega) | \mathcal{F}_\tau^{(m)}]_{x=X_\tau} \\ &= E[g(x, \tau, \tau+h, \omega) | \mathcal{F}_\tau^{(m)}]_{x=X_\tau} = E[g(x, \tau, \tau+h, \omega)]_{x=X_\tau} \\ &= E[f(X_{\tau+h}^{\tau, x})]_{x=X_\tau} = E[f(X_h^{0, x})]_{x=X_\tau} = E[f(F(x, 0, h, \omega))]_{x=X_\tau} . \quad \square \end{aligned}$$

We now extend (7.2.2) to the following:

If $f_1, \dots, f_k$ are bounded Borel functions on $\mathbf{R}^n$, τ an $\mathcal{F}_t^{(m)}$ -stopping time, $\tau < \infty$ a.s. then

$$E^x[f_1(X_{\tau+h_1})f_2(X_{\tau+h_2}) \cdots f_k(X_{\tau+h_k}) | \mathcal{F}_\tau^{(m)}] = E^{X_\tau}[f_1(X_{h_1}) \cdots f_k(X_{h_k})] \quad (7.2.4)$$

for all $0 \leq h_1 \leq h_2 \leq \cdots \leq h_k$. This follows by induction: To illustrate the argument we prove it in the case $k = 2$:

$$\begin{aligned} E^x[f_1(X_{\tau+h_1})f_2(X_{\tau+h_2}) | \mathcal{F}_\tau^{(m)}] &= E^x[E^x[f_1(X_{\tau+h_1})f_2(X_{\tau+h_2}) | \mathcal{F}_{\tau+h_1}^{(m)}] | \mathcal{F}_\tau^{(m)}] \\ &= E^x[f_1(X_{\tau+h_1}) E^x[f_2(X_{\tau+h_2}) | \mathcal{F}_{\tau+h_1}^{(m)}] | \mathcal{F}_\tau^{(m)}] \\ &= E^x[f_1(X_{\tau+h_1}) E^{X_{\tau+h_1}}[f_2(X_{h_2-h_1})] | \mathcal{F}_\tau^{(m)}] \\ &= E^{X_\tau}[f_1(X_{h_1}) E^{X_{h_1}}[f_2(X_{h_2-h_1})]] \\ &= E^{X_\tau}[f_1(X_{h_1}) E^x[f_2(X_{h_2}) | \mathcal{F}_{h_1}^{(m)}]] = E^{X_\tau}[f_1(X_{h_1}) f_2(X_{h_2})] , \quad \text{as claimed} . \end{aligned}$$

Next we proceed to formulate the general version we need: Let $\mathcal{H}$ be the set of all real $\mathcal{M}_\infty$ -measurable functions. For $t \geq 0$ we define the *shift operator*

$$\theta_t: \mathcal{H} \rightarrow \mathcal{H}$$

as follows:

If $\eta = g_1(X_{t_1}) \cdots g_k(X_{t_k})$ (g_i Borel measurable, $t_i \geq 0$) we put

$$\theta_t \eta = g_1(X_{t_1+t}) \cdots g_k(X_{t_k+t}) .$$

Now extend in the natural way to all functions in $\mathcal{H}$ by taking limits of sums of such functions. Then it follows from (7.2.4) that

$$E^x[\theta_\tau \eta | \mathcal{F}_\tau^{(m)}] = E^{X_\tau}[\eta] \quad (7.2.5)$$

for all stopping times τ and all bounded $\eta \in \mathcal{H}$, where

$$(\theta_\tau \eta)(\omega) = (\theta_t \eta)(\omega) \quad \text{if } \tau(\omega) = t .$$

Hitting distribution, harmonic measure and the mean value property

We will apply this to the following situation: Let $H \subset \mathbf{R}^n$ be measurable and let τ_H be the first exit time from H for an Itô diffusion X_t . Let α be another stopping time, g a bounded continuous function on $\mathbf{R}^n$ and put

$$\eta = g(X_{\tau_H})\mathcal{X}_{\{\tau_H < \infty\}}, \quad \tau_H^\alpha = \inf\{t > \alpha; X_t \notin H\}.$$

Then we have

$$\theta_\alpha \eta \cdot \mathcal{X}_{\{\alpha < \infty\}} = g(X_{\tau_H^\alpha})\mathcal{X}_{\{\tau_H^\alpha < \infty\}}. \quad (7.2.6)$$

To prove (7.2.6) we approximate η by functions $\eta^{(k)}$; $k = 1, 2, \dots$, of the form

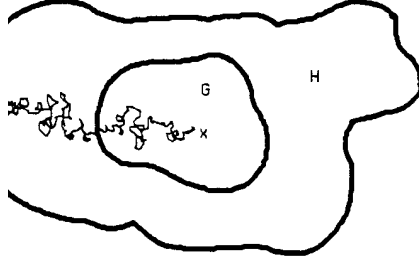
$$\eta^{(k)} = \sum_j g(X_{t_j})\mathcal{X}_{[t_j, t_{j+1})}(\tau_H), \quad t_j = j \cdot 2^{-k}, \quad j = 0, 1, 2, \dots$$

Now

$$\begin{aligned} \theta_t \mathcal{X}_{[t_j, t_{j+1})}(\tau_H) &= \theta_t \mathcal{X}_{\{\forall r \in (0, t_j) X_r \in H \& \exists s \in [t_j, t_{j+1}) X_s \notin H\}} \\ &= \mathcal{X}_{\{\forall r \in (0, t_j) X_{r+t} \in H \& \exists s \in [t_j, t_{j+1}) X_{s+t} \notin H\}} \\ &= \mathcal{X}_{\{\forall u \in (t, t_j+t) X_u \in H \& \exists v \in [t_j+t, t_{j+1}+t) X_v \notin H\}} = \mathcal{X}_{[t_j+t, t_{j+1}+t)}(\tau_H^t). \end{aligned}$$

So we see that

$$\begin{aligned} \theta_t \eta &= \lim_k \theta_t \eta^{(k)} = \lim_k \sum_j g(X_{t_j+t})\mathcal{X}_{[t_j+t, t_{j+1}+t)}(\tau_H^t) \\ &= g(X_{\tau_H^t}) \cdot \mathcal{X}_{\{\tau_H^t < \infty\}}, \quad \text{which is (7.2.6).} \end{aligned}$$



In particular, if $\alpha = \tau_G$ with $G \subset\subset H$ measurable, $\tau_H < \infty$ a.s. Q^x , then we have $\tau_H^\alpha = \tau_H$ and so

$$\theta_{\tau_G} g(X_{\tau_H}) = g(X_{\tau_H}). \quad (7.2.7)$$

So if f is any bounded measurable function we obtain from (7.2.5) and (7.2.7):

$$E^x[f(X_{\tau_H})] = E^x[E^{X_{\tau_G}}[f(X_{\tau_H})]] = \int_{\partial G} E^y[f(X_{\tau_H})] \cdot Q^x[X_{\tau_G} \in dy] \quad (7.2.8)$$

for $x \in G$.

(Define $\mu_H^x(F) = Q^x(X_{\tau_H} \in F)$ and approximate f in $L^1(\mu_H^x)$ by continuous functions g satisfying (7.2.7)). In other words, the expected value of f at X_{τ_H} when starting at $x \in G$ can be obtained by integrating the expected value when starting at $y \in \partial G$ with respect to the *hitting distribution* (“harmonic measure”) of X on ∂G . This can be restated as follows:

Define **the harmonic measure of X on ∂G** , μ_G^x , by

$$\mu_G^x(F) = Q^x[X_{\tau_G} \in F] \quad \text{for } F \subset \partial G, x \in G.$$

Then the function

$$\phi(x) = E^x[f(X_{\tau_H})]$$

satisfies **the mean value property**:

$$\phi(x) = \int_{\partial G} \phi(y) d\mu_G^x(y), \quad \text{for all } x \in G \quad (7.2.9)$$

for all Borel sets $G \subset \subset H$.

This is an important ingredient in our solution of the generalized Dirichlet problem in Chapter 9.

7.3 The Generator of an Itô Diffusion

It is fundamental for many applications that we can associate a second order partial differential operator A to an Itô diffusion X_t . The basic connection between A and X_t is that A is the *generator* of the process X_t :

Definition 7.3.1. Let $\{X_t\}$ be a (time-homogeneous) Itô diffusion in $\mathbf{R}^n$. The (infinitesimal) generator A of X_t is defined by

$$Af(x) = \lim_{t \downarrow 0} \frac{E^x[f(X_t)] - f(x)}{t}; \quad x \in \mathbf{R}^n.$$

The set of functions $f: \mathbf{R}^n \rightarrow \mathbf{R}$ such that the limit exists at x is denoted by $\mathcal{D}_A(x)$, while $\mathcal{D}_A$ denotes the set of functions for which the limit exists for all $x \in \mathbf{R}^n$.

To find the relation between A and the coefficients b, σ in the stochastic differential equation (7.1.4) defining X_t we need the following result, which is useful in many connections:

Lemma 7.3.2. *Let $Y_t = Y_t^x$ be an Itô process in $\mathbf{R}^n$ of the form*

$$Y_t^x(\omega) = x + \int_0^t u(s, \omega) ds + \int_0^t v(s, \omega) dB_s(\omega)$$

where B is m -dimensional. Let $f \in C_0^2(\mathbf{R}^n)$, i.e. $f \in C^2(\mathbf{R}^n)$ and f has compact support, and let τ be a stopping time with respect to $\{\mathcal{F}_t^{(m)}\}$, and assume that $E^x[\tau] < \infty$. Assume that $u(t, \omega)$ and $v(t, \omega)$ are bounded on the set of (t, ω) such that $Y(t, \omega)$ belongs to the support of f . Then

$$E^x[f(Y_\tau)] =$$

$$f(x) + E^x \left[\int_0^\tau \left(\sum_i u_i(s, \omega) \frac{\partial f}{\partial x_i}(Y_s) + \frac{1}{2} \sum_{i,j} (vv^T)_{i,j}(s, \omega) \frac{\partial^2 f}{\partial x_i \partial x_j}(Y_s) \right) ds \right],$$

where E^x is the expectation w.r.t. the natural probability law R^x for Y_t starting at x :

$$R^x[Y_{t_1} \in F_1, \dots, Y_{t_k} \in F_k] = P^0[Y_{t_1}^x \in F_1, \dots, Y_{t_k}^x \in F_k], \quad F_i \text{ Borel sets}.$$

Proof. Put $Z = f(Y)$ and apply Itô's formula (To simplify the notation we suppress the index t and let $Y_1, \dots, Y_n$ and $B_1, \dots, B_m$ denote the coordinates of Y and B , respectively)

$$\begin{aligned} dZ &= \sum_i \frac{\partial f}{\partial x_i}(Y) dY_i + \frac{1}{2} \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j}(Y) dY_i dY_j \\ &= \sum_i u_i \frac{\partial f}{\partial x_i} dt + \frac{1}{2} \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} (v dB)_i (v dB)_j + \sum_i \frac{\partial f}{\partial x_i} (v dB)_i. \end{aligned}$$

Since

$$\begin{aligned} (v dB)_i \cdot (v dB)_j &= \left(\sum_k v_{ik} dB_k \right) \left(\sum_n v_{jn} dB_n \right) \\ &= \left(\sum_k v_{ik} v_{jk} \right) dt = (vv^T)_{ij} dt, \end{aligned}$$

this gives

$$\begin{aligned} f(Y_t) &= f(Y_0) + \int_0^t \left(\sum_i u_i \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i,j} (vv^T)_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} \right) ds \\ &\quad + \sum_{i,k} \int_0^t v_{ik} \frac{\partial f}{\partial x_i} dB_k. \end{aligned} \tag{7.3.1}$$

Hence

$$\begin{aligned} E^x[f(Y_\tau)] &= f(x) + E^x \left[\int_0^\tau \left(\sum_i u_i \frac{\partial f}{\partial x_i}(Y) + \frac{1}{2} \sum_{i,j} (vv^T)_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j}(Y) \right) ds \right] \\ &\quad + \sum_{i,k} E^x \left[\int_0^\tau v_{ik} \frac{\partial f}{\partial x_i}(Y) dB_k \right]. \end{aligned} \quad (7.3.2)$$

If g is a bounded Borel function, $|g| \leq M$ say, then for all integers k we have

$$E^x \left[\int_0^{\tau \wedge k} g(Y_s) dB_s \right] = E^x \left[\int_0^k \mathcal{X}_{\{s < \tau\}} g(Y_s) dB_s \right] = 0,$$

since $g(Y_s)$ and $\mathcal{X}_{\{s < \tau\}}$ are both $\mathcal{F}_s^{(m)}$ -measurable. Moreover

$$\begin{aligned} E^x \left[\left(\int_0^\tau g(Y_s) dB_s - \int_0^{\tau \wedge k} g(Y_s) dB_s \right)^2 \right] &= E^x \left[\int_{\tau \wedge k}^\tau g^2(Y_s) ds \right] \\ &\leq M^2 E^x[\tau - \tau \wedge k] \rightarrow 0. \end{aligned}$$

Therefore

$$0 = \lim_{k \rightarrow \infty} E^x \left[\int_0^{\tau \wedge k} g(Y_s) dB_s \right] = E^x \left[\int_0^\tau g(Y_s) dB_s \right].$$

Combining this with (7.3.2) we get Lemma 7.3.2. $\square$

This gives immediately the formula for the generator A of an Itô diffusion:

Theorem 7.3.3. *Let X_t be the Itô diffusion*

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t.$$

If $f \in C_0^2(\mathbf{R}^n)$ then $f \in \mathcal{D}_A$ and

$$Af(x) = \sum_i b_i(x) \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i,j} (\sigma \sigma^T)_{i,j}(x) \frac{\partial^2 f}{\partial x_i \partial x_j}. \quad (7.3.3)$$

Proof. This follows from Lemma 7.3.2 (with $\tau = t$) and the definition of A . $\square$

Example 7.3.4. The n -dimensional Brownian motion is of course the solution of the stochastic differential equation

$$dX_t = dB_t ,$$

i.e. we have $b = 0$ and $\sigma = I_n$, the n -dimensional identity matrix. So the generator of B_t is

$$Af = \frac{1}{2} \sum \frac{\partial^2 f}{\partial x_i^2} ; \quad f = f(x_1, \dots, x_n) \in C_0^2(\mathbf{R}^n)$$

i.e. $A = \frac{1}{2} \Delta$, where Δ is the Laplace operator.

Example 7.3.5 (The graph of Brownian motion). Let B denote 1-dimensional Brownian motion and let $X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$ be the solution of the stochastic differential equation

$$\begin{cases} dX_1 = dt ; & X_1(0) = t_0 \\ dX_2 = dB ; & X_2(0) = x_0 \end{cases}$$

i.e.

$$dX = bdt + \sigma dB ; \quad X(0) = \begin{pmatrix} t_0 \\ X_0 \end{pmatrix} ,$$

with $b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\sigma = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. In other words, X may be regarded as the graph of Brownian motion. The generator A of X is given by

$$Af = \frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} ; \quad f = f(t, x) \in C_0^2(\mathbf{R}^n) .$$

From now on we will, unless otherwise stated, let $A = A_X$ denote the generator of the Itô diffusion X_t . We let $L = L_X$ denote the differential operator given by the right hand side of (7.3.3). From Theorem 7.3.3 we know that A_X and L_X coincide on $C_0^2(\mathbf{R}^n)$.

7.4 The Dynkin Formula

If we combine (7.3.2) and (7.3.3) we get:

Theorem 7.4.1 (Dynkin's formula).

Let $f \in C_0^2(\mathbf{R}^n)$. Suppose τ is a stopping time, $E^x[\tau] < \infty$. Then

$$E^x[f(X_\tau)] = f(x) + E^x \left[\int_0^\tau Af(X_s) ds \right] . \quad (7.4.1)$$

Remarks.

- (i) Note that if τ is the first exit time of a bounded set, $E^x[\tau] < \infty$, then (7.4.1) holds for any function $f \in C^2$.
- (ii) For a more general version of Theorem 7.4.1 see Dynkin (1965 I), p. 133.

Example 7.4.2. Consider n -dimensional Brownian motion $B = (B_1, \dots, B_n)$ starting at $a = (a_1, \dots, a_n) \in \mathbf{R}^n$ ($n \geq 1$) and assume $|a| < R$. What is the expected value of the first exit time τ_K of B from the ball

$$K = K_R = \{x \in \mathbf{R}^n; |x| < R\} ?$$

Choose an integer k and apply Dynkin's formula with $X = B$, $\tau = \sigma_k = \min(k, \tau_K)$, and $f \in C_0^2$ such that $f(x) = |x|^2$ for $|x| \leq R$:

$$\begin{aligned} E^a[f(B_{\sigma_k})] &= f(a) + E^a \left[\int_0^{\sigma_k} \frac{1}{2} \Delta f(B_s) ds \right] \\ &= |a|^2 + E^a \left[\int_0^{\sigma_k} n \cdot ds \right] = |a|^2 + n \cdot E^a[\sigma_k]. \end{aligned}$$

Hence $E^a[\sigma_k] \leq \frac{1}{n}(R^2 - |a|^2)$ for all k . So letting $k \rightarrow \infty$ we conclude that $\tau_K = \lim \sigma_k < \infty$ a.s. and

$$E^a[\tau_K] = \frac{1}{n}(R^2 - |a|^2). \quad (7.4.2)$$

Next we assume that $n \geq 2$ and $|b| > R$. What is the probability that B starting at b ever hits K ?

Let α_k be the first exit time from the annulus

$$A_k = \{x; R < |x| < 2^k R\}; \quad k = 1, 2, \dots$$

and put

$$T_K = \inf\{t > 0; B_t \in K\}.$$

Let $f = f_{n,k}$ be a C^2 function with compact support such that, if $R \leq |x| \leq 2^k R$,

$$f(x) = \begin{cases} -\log |x| & \text{when } n = 2 \\ |x|^{2-n} & \text{when } n > 2. \end{cases}$$

Then, since $\Delta f = 0$ in A_k , we have by Dynkin's formula

$$E^b[f(B_{\alpha_k})] = f(b) \quad \text{for all } k. \quad (7.4.3)$$

Put

$$p_k = P^b[|B_{\alpha_k}| = R], \quad q_k = P^b[|B_{\alpha_k}| = 2^k R].$$

Let us now consider the two cases $n = 2$ and $n > 2$ separately:

$n = 2$. Then we get from (7.4.3)

$$-\log R \cdot p_k - (\log R + k \cdot \log 2)q_k = -\log |b| \quad \text{for all } k. \quad (7.4.4)$$

This implies that $q_k \rightarrow 0$ as $k \rightarrow \infty$, so that

$$P^b[T_K < \infty] = 1, \quad (7.4.5)$$

i.e. Brownian motion is *recurrent* in $\mathbf{R}^2$. (See Port and Stone (1979)).

$n > 2$. In this case (7.4.3) gives

$$p_k \cdot R^{2-n} + q_k \cdot (2^k R)^{2-n} = |b|^{2-n}.$$

Since $0 \leq q_k \leq 1$ we get by letting $k \rightarrow \infty$

$$\lim_{k \rightarrow \infty} p_k = P^b[T_K < \infty] = \left(\frac{|b|}{R}\right)^{2-n},$$

i.e. Brownian motion is *transient* in $\mathbf{R}^n$ for $n > 2$.

7.5 The Characteristic Operator

We now introduce an operator which is closely related to the generator A , but is more suitable in many situations, for example in the solution of the Dirichlet problem.

Definition 7.5.1. Let $\{X_t\}$ be an Itô diffusion. The characteristic operator $\mathcal{A} = \mathcal{A}_X$ of $\{X_t\}$ is defined by

$$\mathcal{A}f(x) = \lim_{U \downarrow x} \frac{E^x[f(X_{\tau_U})] - f(x)}{E^x[\tau_U]}, \quad (7.5.1)$$

where the U 's are open sets U_k decreasing to the point x , in the sense that $U_{k+1} \subset U_k$ and $\bigcap_k U_k = \{x\}$, and $\tau_U = \inf\{t > 0; X_t \notin U\}$ is the first exit time from U for X_t . The set of functions f such that the limit (7.5.1) exists for all $x \in \mathbf{R}^n$ (and all $\{U_k\}$) is denoted by $\mathcal{D}_\mathcal{A}$. If $E^x[\tau_U] = \infty$ for all open $U \ni x$, we define $\mathcal{A}f(x) = 0$.

It turns out that $\mathcal{D}_A \subseteq \mathcal{D}_\mathcal{A}$ always and that

$$Af = \mathcal{A}f \quad \text{for all } f \in \mathcal{D}_\mathcal{A}.$$

(See Dynkin (1965 I, p. 143).)

We will only need that $\mathcal{A}_X$ and L_X coincide on C^2 . To obtain this we first clarify a property of exit times.

Definition 7.5.2. A point $x \in \mathbf{R}^n$ is called a trap for $\{X_t\}$ if

$$Q^x(\{X_t = x \text{ for all } t\}) = 1.$$

In other words, x is trap if and only if $\tau_{\{x\}} = \infty$ a.s. Q^x . For example, if $b(x_0) = \sigma(x_0) = 0$, then x_0 is a trap for X_t (by strong uniqueness of X_t).

Lemma 7.5.3. If x is not a trap for X_t , then there exists an open set $U \ni x$ such that

$$E^x[\tau_U] < \infty.$$

Proof. See Lemma 5.5 p. 139 in Dynkin (1965 I).

Theorem 7.5.4. Let $f \in C^2$. Then $f \in \mathcal{D}_A$ and

$$\mathcal{A}f = \sum_i b_i \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i,j} (\sigma \sigma^T)_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j}. \quad (7.5.2)$$

Proof. As before we let L denote the operator defined by the right hand side of (7.5.2). If x is a trap for $\{X_t\}$ then $\mathcal{A}f(x) = 0$. Choose a bounded open set V such that $x \in V$. Modify f to f_0 outside V such that $f_0 \in C_0^2(\mathbf{R}^n)$. Then $f_0 \in \mathcal{D}_A(x)$ and $0 = \mathcal{A}f_0(x) = Lf_0(x) = Lf(x)$. Hence $\mathcal{A}f(x) = Lf(x) = 0$ in this case. If x is not a trap, choose a bounded open set $U \ni x$ such that $E^x[\tau_U] < \infty$. Then by Dynkin's formula (Theorem 7.4.1) (and the following Remark (i)), writing $\tau_U = \tau$

$$\begin{aligned} \left| \frac{E^x[f(X_\tau)] - f(x)}{E^x[\tau]} - Lf(x) \right| &= \frac{|E^x[\int_0^\tau \{(Lf)(X_s) - Lf(x)\} ds]|}{E^x[\tau]} \\ &\leq \sup_{y \in U} |Lf(x) - Lf(y)| \rightarrow 0 \quad \text{as } U \downarrow x, \end{aligned}$$

since Lf is a continuous function.

Remark. We have now obtained that an Itô diffusion is a continuous, strong Markov process such that the domain of definition of its characteristic operator includes C^2 . Thus an Itô diffusion is a *diffusion* in the sense of Dynkin (1965 I).

Example 7.5.5 (Brownian motion on the unit circle). The characteristic operator of the process $Y = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$ from Example 5.1.4 satisfying the stochastic differential equations (5.1.13), i.e.

$$\begin{cases} dY_1 = -\frac{1}{2}Y_1 dt - Y_2 dB \\ dY_2 = -\frac{1}{2}Y_2 dt + Y_1 dB \end{cases}$$

is

$$\mathcal{A}f(y_1, y_2) = \frac{1}{2} \left[y_2^2 \frac{\partial^2 f}{\partial y_1^2} - 2y_1 y_2 \frac{\partial^2 f}{\partial y_1 \partial y_2} + y_1^2 \frac{\partial^2 f}{\partial y_2^2} - y_1 \frac{\partial f}{\partial y_1} - y_2 \frac{\partial f}{\partial y_2} \right].$$

This is because $dY = -\frac{1}{2}Ydt + KYdB$, where

$$K = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

so that

$$dY = b(Y)dt + \sigma(Y)dB$$

with

$$b(y_1, y_2) = \begin{pmatrix} -\frac{1}{2}y_1 \\ -\frac{1}{2}y_2 \end{pmatrix}, \quad \sigma(y_1, y_2) = \begin{pmatrix} -y_2 \\ y_1 \end{pmatrix}$$

and

$$a = \frac{1}{2}\sigma\sigma^T = \frac{1}{2} \begin{pmatrix} y_2^2 & -y_1 y_2 \\ -y_1 y_2 & y_1^2 \end{pmatrix}.$$

Example 7.5.6. Let D be an open subset of $\mathbf{R}^n$ such that $\tau_D < \infty$ a.s. Q^x for all x . Let ϕ be a bounded, measurable function on ∂D and define

$$\tilde{\phi}(x) = E^x[\phi(X_{\tau_D})]$$

($\tilde{\phi}$ is called the *X-harmonic extension of ϕ*). Then if U is open, $x \in U \subset\subset D$, we have by (7.2.8) that

$$E^x[\tilde{\phi}(X_{\tau_U})] = E^x[E^{X_{\tau_U}}[\phi(X_{\tau_D})]] = E^x[\phi(X_{\tau_D})] = \tilde{\phi}(x).$$

So $\tilde{\phi} \in \mathcal{D}_A$ and

$$\mathcal{A}\tilde{\phi} = 0 \quad \text{in } D,$$

in spite of the fact that in general $\tilde{\phi}$ need not even be continuous in D (See Example 9.2.1).

Exercises

7.1. Find the generator of the following Itô diffusions:

- a) $dX_t = \mu X_t dt + \sigma dB_t$ (The Ornstein-Uhlenbeck process) ($B_t \in \mathbf{R}$; μ, σ constants).
- b) $dX_t = r X_t dt + \alpha X_t dB_t$ (The geometric Brownian motion) ($B_t \in \mathbf{R}$; r, α constants).
- c) $dY_t = r dt + \alpha Y_t dB_t$ ($B_t \in \mathbf{R}$; r, α constants)

- d) $dY_t = \begin{bmatrix} dt \\ dX_t \end{bmatrix}$ where X_t is as in a)
- e) $\begin{bmatrix} dX_1 \\ dX_2 \end{bmatrix} = \begin{bmatrix} 1 \\ X_2 \end{bmatrix} dt + \begin{bmatrix} 0 \\ e^{X_1} \end{bmatrix} dB_t \quad (B_t \in \mathbf{R})$
- f) $\begin{bmatrix} dX_1 \\ dX_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} dt + \begin{bmatrix} 1 & 0 \\ 0 & X_1 \end{bmatrix} \begin{bmatrix} dB_1 \\ dB_2 \end{bmatrix}$
- g) $\bar{X}(t) = (X_1, X_2, \dots, X_n)$, where

$$dX_k(t) = r_k X_k dt + X_k \cdot \sum_{j=1}^n \alpha_{kj} dB_j ; \quad 1 \leq k \leq n$$

(($B_1, \dots, B_n$) is Brownian motion in $\mathbf{R}^n$, r_k and α_{kj} are constants).

7.2. Find an Itô diffusion (i.e. write down the stochastic differential equation for it) whose generator is the following:

- a) $Af(x) = f'(x) + f''(x); \quad f \in C_0^2(\mathbf{R})$
- b) $Af(t, x) = \frac{\partial f}{\partial t} + cx \frac{\partial f}{\partial x} + \frac{1}{2} \alpha^2 x^2 \frac{\partial^2 f}{\partial x^2}; \quad f \in C_0^2(\mathbf{R}^2)$,
where c, α are constants.
- c) $Af(x_1, x_2) = 2x_2 \frac{\partial f}{\partial x_1} + \ln(1 + x_1^2 + x_2^2) \frac{\partial f}{\partial x_2}$
 $+ \frac{1}{2}(1 + x_1^2) \frac{\partial^2 f}{\partial x_1^2} + x_1 \frac{\partial^2 f}{\partial x_1 \partial x_2} + \frac{1}{2} \cdot \frac{\partial^2 f}{\partial x_2^2}; \quad f \in C_0^2(\mathbf{R}^2).$

7.3. Let B_t be Brownian motion on $\mathbf{R}$, $B_0 = 0$ and define

$$X_t = X_t^x = x \cdot e^{ct + \alpha B_t},$$

where c, α are constants. Prove directly from the definition that X_t is a Markov process.

7.4. Let B_t^x be 1-dimensional Brownian motion starting at $x \in \mathbf{R}^+$. Put

$$\tau = \inf\{t > 0; B_t^x = 0\}.$$

- a) Prove that $\tau < \infty$ a.s. P^x for all $x > 0$. (Hint: See Example 7.4.2, second part).
- b) Prove that $E^x[\tau] = \infty$ for all $x > 0$. (Hint: See Example 7.4.2, first part).
- 7.5.** Let the functions b, σ satisfy condition (5.2.1) of Theorem 5.2.1, with a constant C independent of t , i.e.

$$|b(t, x)| + |\sigma(t, x)| \leq C(1 + |x|) \quad \text{for all } x \in \mathbf{R}^n \text{ and all } t \geq 0.$$

Let X_t be a solution of

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t.$$

Show that

$$E[|X_t|^2] \leq (1 + E[|X_0|^2])e^{Kt} - 1$$

for some constant K independent of t .

(Hint: Use Dynkin's formula with $f(x) = |x|^2$ and $\tau = t \wedge \tau_R$, where $\tau_R = \inf \{t > 0; |X_t| \geq R\}$, and let $R \rightarrow \infty$ to achieve the inequality

$$E[|X_t|^2] \leq E[|X_0|^2] + K \cdot \int_0^t (1 + E[|X_s|^2])ds,$$

which is of the form (5.2.9).)

7.6. Let $g(x, \omega) = f \circ F(x, t, t + h, \omega)$ be as in the proof of Theorem 7.1.2. Assume that f is continuous.

a) Prove that the map $x \rightarrow g(x, \cdot)$ is continuous from $\mathbf{R}^n$ into $L^2(P)$ by using (5.2.9).

For simplicity assume that $n = 1$ in the following.

b) Use a) to prove that $(x, \omega) \rightarrow g(x, \omega)$ is measurable. (Hint: For each $m = 1, 2, \dots$ put $\xi_k = \xi_k^{(m)} = k \cdot 2^{-m}$, $k = 1, 2, \dots$. Then

$$g^{(m)}(x, \cdot) := \sum_k g(\xi_k, \cdot) \cdot \mathcal{X}_{\{\xi_k \leq x < \xi_{k+1}\}}$$

converges to $g(x, \cdot)$ in $L^2(P)$ for each x . Deduce that $g^{(m)} \rightarrow g$ in $L^2(dm_R \times dP)$ for all R , where dm_R is Lebesgue measure on $\{|x| \leq R\}$. So a subsequence of $g^{(m)}(x, \omega)$ converges to $g(x, \omega)$ for a.a. (x, ω) .)

7.7. Let B_t be Brownian motion on $\mathbf{R}^n$ starting at $x \in \mathbf{R}^n$ and let $D \subset \mathbf{R}^n$ be an open ball centered at x .

a) Use Exercise 2.15 to prove that the harmonic measure μ_D^x of B_t is rotation invariant (about x) on the sphere ∂D . Conclude that μ_D^x coincides with normalized surface measure σ on ∂D .

b) Let ϕ be a bounded measurable function on a bounded open set $W \subset \mathbf{R}^n$ and define

$$u(x) = E^x[\phi(B_{\tau_W})] \quad \text{for } x \in W.$$

Prove that u satisfies the classical mean value property:

$$u(x) = \int_{\partial D} u(y) d\sigma(y)$$

for all balls D centered at x with $\overline{D} \subset W$.

- 7.8.** Let $\{\mathcal{N}_t\}$ be a right-continuous family of σ -algebras of subsets of Ω , containing all sets of measure zero.
- a) Let τ_1, τ_2 be stopping times (w.r.t. $\mathcal{N}_t$). Prove that $\tau_1 \wedge \tau_2$ and $\tau_1 \vee \tau_2$ are stopping times.
 - b) If $\{\tau_n\}$ is a decreasing family of stopping times prove that $\tau := \lim_n \tau_n$ is a stopping time.
 - c) If X_t is an Itô diffusion in $\mathbf{R}^n$ and $F \subset \mathbf{R}^n$ is closed, prove that τ_F is a stopping time w.r.t. $\mathcal{M}_t$. (Hint: Consider open sets decreasing to F).
- 7.9.** Let X_t be a geometric Brownian motion, i.e.

$$dX_t = rX_t dt + \alpha X_t dB_t, \quad X_0 = x > 0$$

where $B_t \in \mathbf{R}$; r, α are constants.

- a) Find the generator A of X_t and compute $Af(x)$ when $f(x) = x^\gamma$; $x > 0$, γ constant.
- b) If $r < \frac{1}{2}\alpha^2$ then $X_t \rightarrow 0$ as $t \rightarrow \infty$, a.s. Q^x (Example 5.1.1). But what is the probability p that X_t , when starting from $x < R$, ever hits the value R ? Use Dynkin's formula with $f(x) = x^{\gamma_1}$, $\gamma_1 = 1 - \frac{2r}{\alpha^2}$, to prove that

$$p = \left(\frac{x}{R}\right)^{\gamma_1}.$$

- c) If $r > \frac{1}{2}\alpha^2$ then $X_t \rightarrow \infty$ as $t \rightarrow \infty$, a.s. Q^x . Put

$$\tau = \inf\{t > 0; X_t \geq R\}.$$

Use Dynkin's formula with $f(x) = \ln x$, $x > 0$ to prove that

$$E^x[\tau] = \frac{\ln \frac{R}{x}}{r - \frac{1}{2}\alpha^2}.$$

(Hint: First consider exit times from (ρ, R) , $\rho > 0$ and then let $\rho \rightarrow 0$. You need estimates for

$$(1 - p(\rho)) \ln \rho,$$

where

$$p(\rho) = Q^x[X_t \text{ reaches the value } R \text{ before } \rho],$$

which you can get from the calculations in a), b).)

7.10. Let X_t be the geometric Brownian motion

$$dX_t = rX_t dt + \alpha X_t dB_t .$$

Find $E^x[X_T|\mathcal{F}_t]$ for $t \leq T$ by

- a) using the Markov property
- and
- b) writing $X_t = x e^{rt} M_t$, where

$$M_t = \exp(\alpha B_t - \frac{1}{2}\alpha^2 t) \quad \text{is a martingale .}$$

7.11. Let X_t be an Itô diffusion in $\mathbf{R}^n$ and let $f: \mathbf{R}^n \rightarrow \mathbf{R}$ be a function such that

$$E^x \left[\int_0^\infty |f(X_t)| dt \right] < \infty \quad \text{for all } x \in \mathbf{R}^n .$$

Let τ be a stopping time. Use the strong Markov property to prove that

$$E^x \left[\int_\tau^\infty f(X_t) dt \right] = E^x[g(X_\tau)] ,$$

where

$$g(y) = E^y \left[\int_0^\infty f(X_t) dt \right] .$$

7.12. (Local martingales)

An $\mathcal{N}_t$ -adapted stochastic process $Z(t) \in \mathbf{R}^n$ is called a *local martingale* with respect to the given filtration $\{\mathcal{N}_t\}$ if there exists an increasing sequence of $\mathcal{N}_t$ -stopping times τ_k such that

$$\tau_k \rightarrow \infty \quad \text{a.s. as } k \rightarrow \infty$$

and

$$Z(t \wedge \tau_k) \quad \text{is an } \mathcal{N}_t\text{-martingale for all } k .$$

- a) Show that if $Z(t)$ is a local martingale and there exists a constant $T \leq \infty$ such that the family $\{Z(\tau)\}_{\tau \leq T}$ is uniformly integrable (Appendix C) then $\{Z(t)\}_{t \leq T}$ is a martingale.
- b) In particular, if $Z(t)$ is a local martingale and there exists a constant $K < \infty$ such that

$$E[Z^2(\tau)] \leq K$$

for all stopping times $\tau \leq T$, then $\{Z(t)\}_{t \leq T}$ is a martingale.

- c) Show that if $Z(t)$ is a *lower bounded* local martingale, then $Z(t)$ is a supermartingale (Appendix C).

7.13. a) Let $B_t \in \mathbf{R}^2$, $B_0 = x \neq 0$. Fix $0 < \epsilon < R < \infty$ and define

$$X_t = \ln |B_{t \wedge \tau}|; \quad t \geq 0$$

where

$$\tau = \inf \{t > 0; |B_t| \leq \epsilon \text{ or } |B_t| \geq R\}.$$

Prove that X_t is an $\mathcal{F}_{t \wedge \tau}$ -martingale. (Hint: Use Exercise 4.8.)

Deduce that $\ln |B_t|$ is a local martingale (Exercise 7.12).

b) Let $B_t \in \mathbf{R}^n$ for $n \geq 3$, $B_0 = x \neq 0$. Fix $\epsilon > 0$, $R < \infty$ and define

$$Y_t = |B_{t \wedge \tau}|^{2-n}; \quad t \geq 0$$

where

$$\tau = \inf \{t > 0; |B_t| \leq \epsilon \text{ or } |B_t| \geq R\}.$$

Prove that Y_t is an $\mathcal{F}_{t \wedge \tau}$ -martingale.

Deduce that $|B_t|^{2-n}$ is a local martingale.

7.14. (Doob's h -transform)

Let B_t be n -dimensional Brownian motion, $D \subset \mathbf{R}^n$ a bounded open set and $h > 0$ a harmonic function on D (i.e. $\Delta h = 0$ in D). Let X_t be the solution of the stochastic differential equation

$$dX_t = \nabla(\ln h)(X_t)dt + dB_t$$

More precisely, choose an increasing sequence $\{D_k\}$ of open subsets of D such that $\overline{D_k} \subset D$ and $\bigcup_{k=1}^{\infty} D_k = D$. Then for each k the equation above can be solved (strongly) for $t < \tau_{D_k}$. This gives in a natural way a solution for $t < \tau := \lim_{k \rightarrow \infty} \tau_{D_k}$.

a) Show that the generator A of X_t satisfies

$$Af = \frac{\Delta(hf)}{2h} \quad \text{for } f \in C_0^2(D).$$

In particular, if $f = \frac{1}{h}$ then $Af = 0$.

b) Use a) to show that if there exists $x_0 \in \partial D$ such that

$$\lim_{x \rightarrow y \in \partial D} h(x) = \begin{cases} 0 & \text{if } y \neq x_0 \\ \infty & \text{if } y = x_0 \end{cases}$$

(i.e. h is a *kernel function*), then

$$\lim_{t \rightarrow \tau} X_t = x_0 \text{ a.s.}$$

(Hint: Consider $E^x[f(X_T)]$ for suitable stopping times T and with $f = \frac{1}{h}$)

In other words, we have imposed a drift on B_t which causes the process to exit from D at the point x_0 only. This can also be formulated as follows: X_t is obtained by *conditioning* B_t to exit from D at x_0 . See Doob (1984).

7.15. Let B_t be 1-dimensional and define

$$F(\omega) = (B_T(\omega) - K)^+$$

where $K > 0$, $T > 0$ are constants.

By the Itô representation theorem (Theorem 4.3.3) we know that there exists $\phi \in \mathcal{V}(0, T)$ such that

$$F(\omega) = E[F] + \int_0^T \phi(t, \omega) dB_t .$$

How do we find ϕ explicitly? This problem is of interest in mathematical finance, where ϕ may be regarded as the replicating portfolio for the contingent claim F (see Chapter 12). Using the Clark-Ocone formula (see Karatzas and Ocone (1991) or Øksendal (1996)) one can deduce that

$$\phi(t, \omega) = E[\mathcal{X}_{[K, \infty)}(B_T) | \mathcal{F}_t] ; \quad t < T . \quad (7.5.3)$$

Use (7.5.3) and the Markov property of Brownian motion to prove that for $t < T$ we have

$$\phi(t, \omega) = \frac{1}{\sqrt{2\pi(T-t)}} \int_K^\infty \exp\left(-\frac{(x - B_t(\omega))^2}{2(T-t)}\right) dx . \quad (7.5.4)$$

7.16. Let B_t be 1-dimensional and let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a bounded function. Prove that if $t < T$ then

$$E^x[f(B_T) | \mathcal{F}_t] = \frac{1}{\sqrt{2\pi(T-t)}} \int_{\mathbf{R}} f(x) \exp\left(-\frac{(x - B_t(\omega))^2}{2(T-t)}\right) dx . \quad (7.5.5)$$

(Compare with (7.5.4).)

7.17. Let B_t be 1-dimensional and put

$$X_t = (x^{1/3} + \frac{1}{3}B_t)^3 ; \quad t \geq 0 .$$

Then we have seen in Exercise 4.15 that X_t is a solution of the stochastic differential equation

$$dX_t = \frac{1}{3}X_t^{1/3}dt + X_t^{2/3}dB_t ; \quad X_0 = x . \quad (7.5.6)$$

Define

$$\tau = \inf\{t > 0; X_t = 0\}$$

and put

$$Y_t = \begin{cases} X_t & \text{for } t \leq \tau \\ 0 & \text{for } t > \tau. \end{cases}$$

Prove that Y_t is also a (strong) solution of (7.5.6). Why does not this contradict the uniqueness assertion of Theorem 5.2.1?

(Hint: Verify that

$$Y_t = x + \int_0^t \frac{1}{3} Y_s^{1/3} ds + \int_0^t Y_s^{2/3} dB_s$$

for all t by splitting the integrals as follows:

$$\int_0^t = \int_0^{t \wedge \tau} + \int_{t \wedge \tau}^t .)$$

7.18. a) Let

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t; \quad X_0 = x$$

be a 1-dimensional Itô diffusion with characteristic operator $\mathcal{A}$. Let $f \in C^2(\mathbf{R})$ be a solution of the differential equation

$$\mathcal{A}f(x) = b(x)f'(x) + \frac{1}{2}\sigma^2(x)f''(x) = 0; \quad x \in \mathbf{R}. \quad (7.5.7)$$

Let $(a, b) \subset \mathbf{R}$ be an open interval such that $x \in (a, b)$ and put

$$\tau = \inf\{t > 0; X_t \notin (a, b)\}.$$

Assume that $\tau < \infty$ a.s. Q^x and define

$$p = P^x[X_\tau = b].$$

Use Dynkin's formula to prove that if $f(b) \neq f(a)$ then

$$p = \frac{f(x) - f(a)}{f(b) - f(a)}. \quad (7.5.8)$$

In other words, the *harmonic measure* $\mu_{(a,b)}^x$ of X on $\partial(a, b) = \{a, b\}$ is given by

$$\mu_{(a,b)}^x(b) = \frac{f(x) - f(a)}{f(b) - f(a)}, \quad \mu_{(a,b)}^x(a) = \frac{f(b) - f(x)}{f(b) - f(a)}. \quad (7.5.9)$$

b) Now specialize to the process

$$X_t = x + B_t; \quad t \geq 0.$$

Prove that

$$p = \frac{x-a}{b-a}. \quad (7.5.10)$$

c) Find p if

$$X_t = x + \mu t + \sigma B_t; \quad t \geq 0$$

where $\mu, \sigma \in \mathbf{R}$ are nonzero constants.

7.19. Let B_t^x be 1-dimensional Brownian motion starting at $x > 0$. Define

$$\tau = \tau(x, \omega) = \inf\{t > 0; B_t^x(\omega) = 0\}.$$

From Exercise 7.4 we know that

$$\tau < \infty \quad \text{a.s. } P^x \text{ and } E^x[\tau] = \infty.$$

What is the distribution of the random variable $\tau(\omega)$?

a) To answer this, first find the *Laplace transform*

$$g(\lambda) := E^x[e^{-\lambda\tau}] \quad \text{for } \lambda > 0.$$

(Hint: Let $M_t = \exp(-\sqrt{2\lambda} B_t - \lambda t)$. Then

$$\{M_{t \wedge \tau}\}_{t \geq 0} \text{ is a bounded martingale.}$$

[Solution: $g(\lambda) = \exp(-\sqrt{2\lambda} x)$.]

b) To find the density $f(t)$ of τ it suffices to find $f(t) = f(t, x)$ such that

$$\int_0^\infty e^{-\lambda t} f(t) dt = \exp(-\sqrt{2\lambda} x) \quad \text{for all } \lambda > 0$$

i.e. to find the *inverse* Laplace transform of $g(\lambda)$. Verify that

$$f(t, x) = \frac{x}{\sqrt{2\pi t^3}} \exp\left(-\frac{x^2}{2t}\right); \quad t > 0.$$